Probability-Weighted Combinatorial Coverage

Risk-Informed Scenario Testing of Autonomous Systems

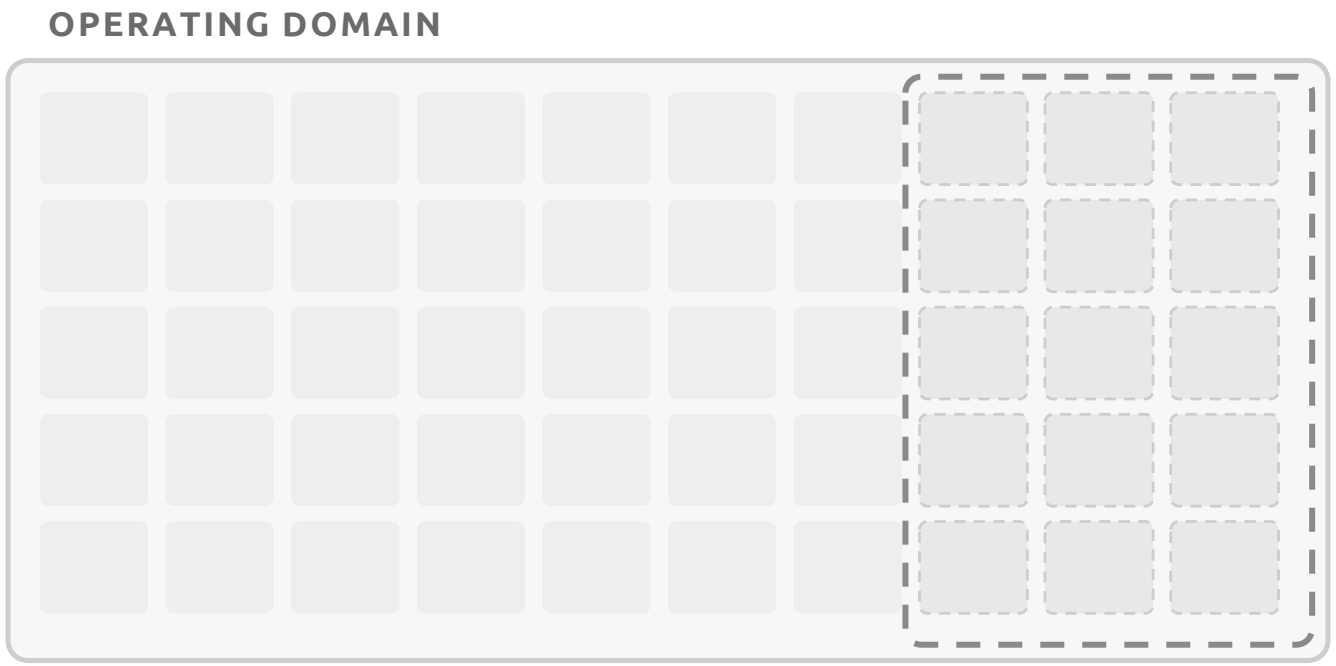
Borhaneddine Hamadou · Chokri Mraidha · Nikolaos Matragkas

Université Paris-Saclay, CEA List · Palaiseau, France

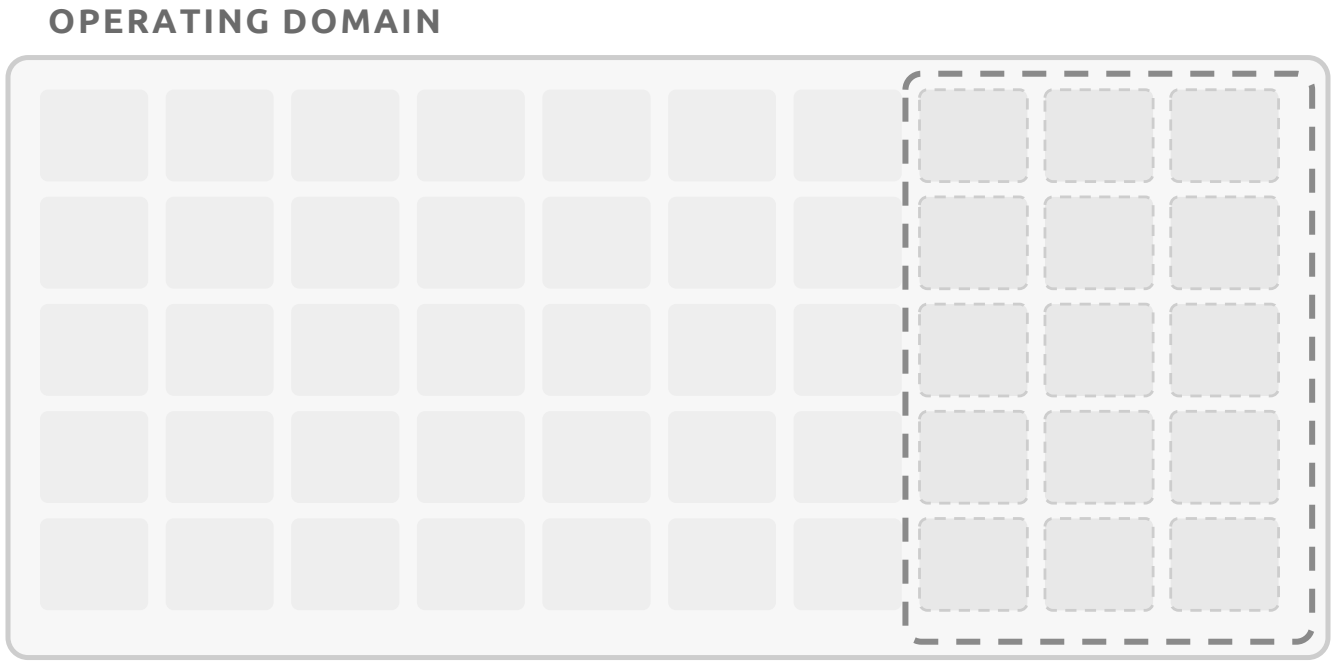


ASE 2026 · 41st IEEE/ACM International Conference on Automated Software Engineering · Munich

We hid part of the operating domain. Then we asked the standard metric: is this suite adequate?



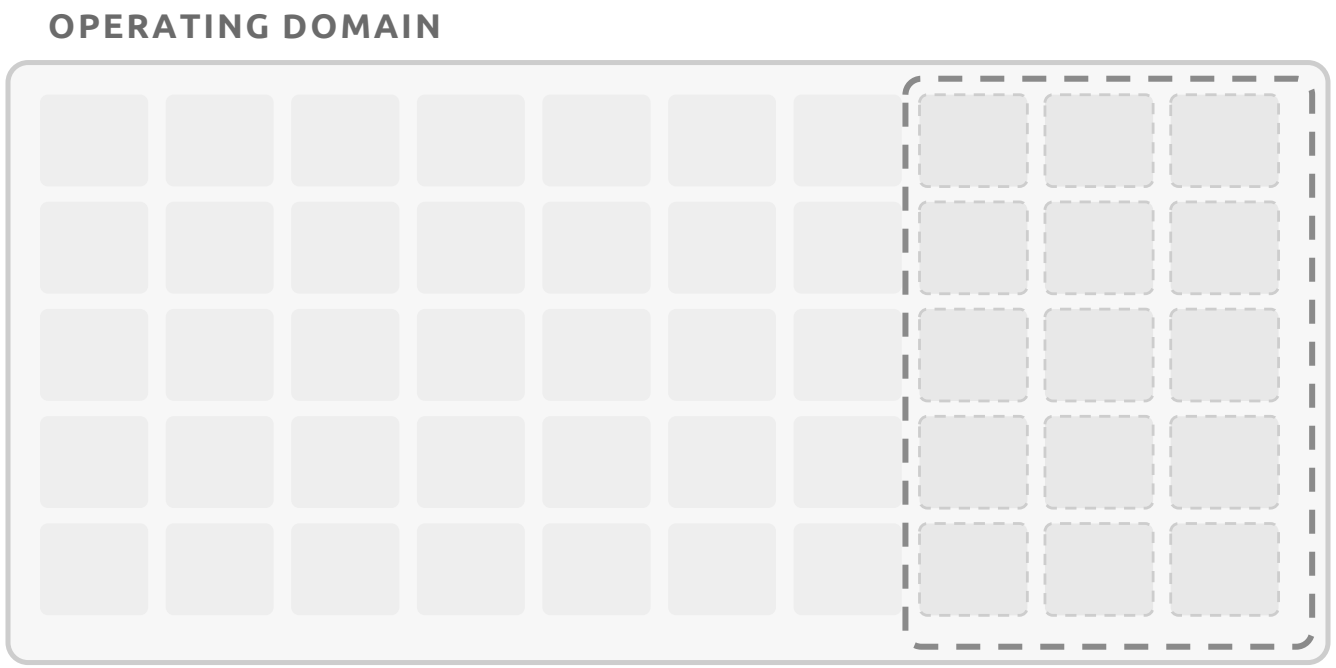
We hid part of the operating domain. Then we asked the standard metric: is this suite adequate?



Is this suite adequate?

asked of standard combinatorial coverage

We hid part of the operating domain. Then we asked the standard metric: is this suite adequate?



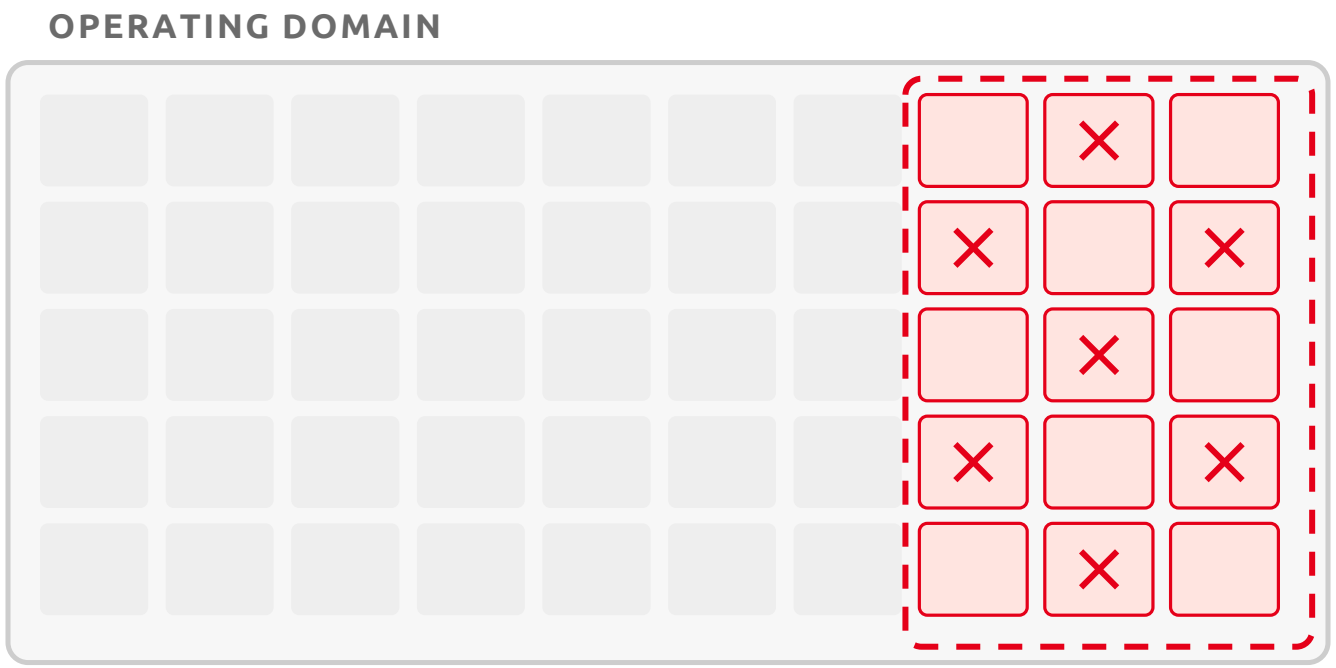
Is this suite adequate?

asked of standard combinatorial coverage

YES

in every split, at every interaction order

We hid part of the operating domain. Then we asked the standard metric: is this suite adequate?



Is this suite adequate?

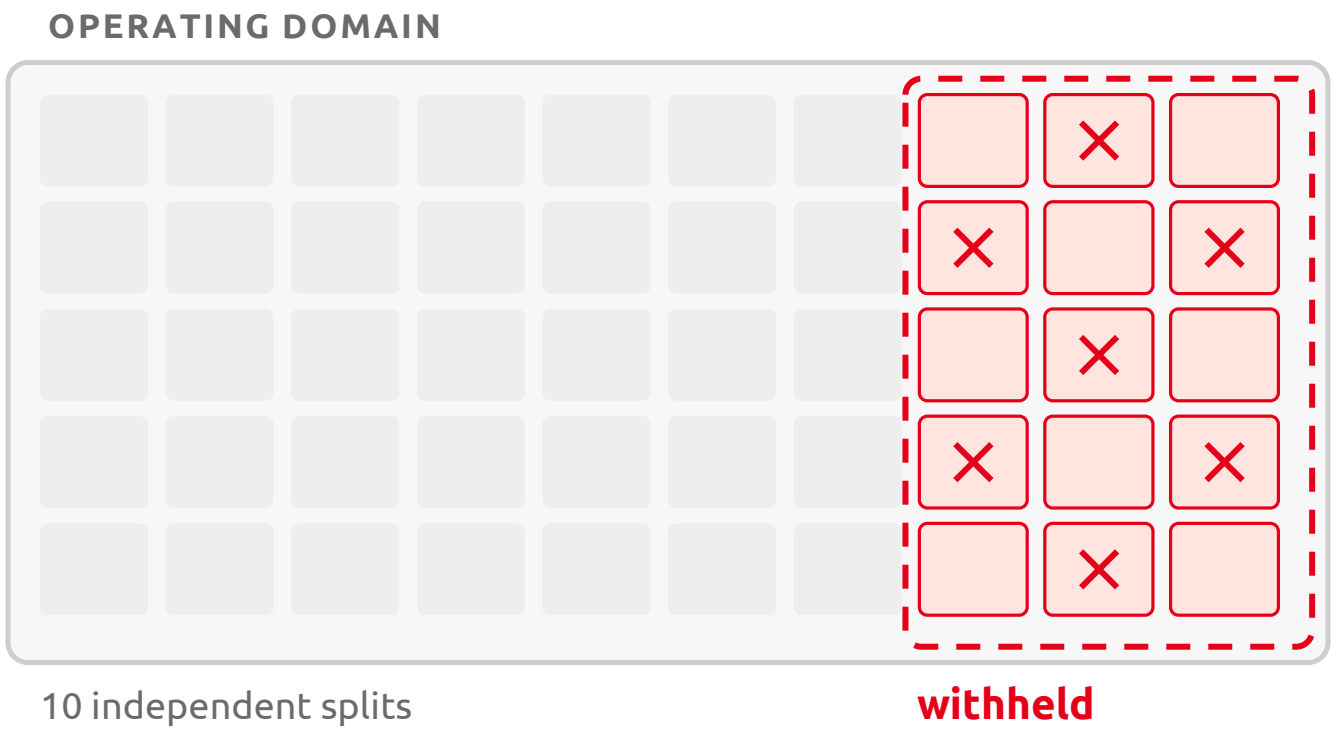
asked of standard combinatorial coverage

YES

in every split, at every interaction order

100% **of the unseen failures**
concealed, in 9 of 10 splits

We hid part of the operating domain. Then we asked the standard metric: is this suite adequate?



Is this suite adequate?

asked of standard combinatorial coverage

YES

in every split, at every interaction order

100% **of the unseen failures** concealed, in 9 of 10 splits

FALSE SUFFICIENCY

THE PROBLEM

Somebody has to decide when testing is enough, and nothing today supports that decision.

SIMULATION CAMPAIGN



scenarios executed

when is it enough?

Stop too early
dangerous conditions never exercised

Never stop
resources spent with no safety gain

ISO 21448 (SOTIF) asks for a *justified* claim that more testing would not substantially increase coverage of safety-relevant conditions.

THE STATE OF THE ART

Today's metrics measure what was tested, all of it equally.

Diversity metrics
geometric spread of test cases in feature space (e.g., TISA [1])

Combinatorial coverage
fraction of t-way parameter combinations exercised [2]



same size, most of them ticked

Both describe what was tested.
Neither says what it was worth testing.

[1] Neelofar Neelofar and Aldeida Aleti. 2024. Towards Reliable AI: Adequacy Metrics for Ensuring the Quality of System-level Testing of Autonomous Vehicles. In Proceedings of the IEEE/ACM 46th International Conference on Software Engineering (Lisbon, Portugal) (ICSE '24). Association for Computing Machinery, New York, NY, USA, Article 68, 12 pages.

[2] Changhai Nie and Hareton Leung. 2011. A survey of combinatorial testing. Comput. Surveys 43, 2, Article 11 (Feb. 2011), 29 pages. doi:10.1145/1883612. 1883618.

THE GAP

A suite can score full marks while missing what the system will **actually meet**.

WHAT THE METRIC COUNTS



100 % covered

WHAT THE SYSTEM ACTUALLY MEETS



same cells, sized by how often they occur

How much of expected real-world operation remains **insufficiently tested**?

THE APPROACH

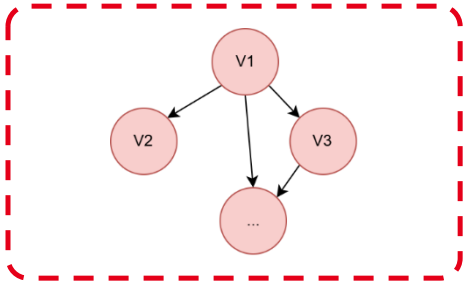
PWCC weights coverage by how often conditions actually occur.



Step 1: learn what the operating world actually looks like.

profiling data

field · fleet · simulation · weather archives



Probabilistic model (e,g, Bayesian Network)

combination of conditions	share of operation
heavy rain + wet road	2 %
heavy rain + ice	0.2 %

the model gives non-zero probability to plausible combinations the data never recorded

STEP 2 · THE STANDARD

Step 2: a condition you meet all the time is held to a stricter standard.

tolerable event rate = risk budget ÷ share of operation

	combination	share of operation	tolerable rate
	heavy rain + wet road	**2 %**	**0.5 events / hour**
	heavy rain + ice	**0.2 %**	**5 events / hour**

the risk budget is your safety requirement, not a tuning knob

STEP 3 · THE TIME

Step 3: a stricter standard takes longer to check.

required exposure = time to be 95 % confident a fault at that rate would have shown

	pair of conditions	share	tolerable rate	required exposure
	heavy rain + wet road	2 %	0.5 / h	**about 6 hours**
	heavy rain + ice	0.2 %	5 / h	**about 36 minutes**

testing effort concentrates where operation actually is

$$1 - e^{-\lambda_{\max}(c)\,E} \geq 1 - \alpha \implies E \geq \frac{-\ln(\alpha)}{\lambda_{\max}(c)}$$

1 2 **3** 4

STEP 4 · THE STOP

Step 4: stop when under 5% of expected operation is still insufficiently tested.

Score = share of operation

that reached its required exposure

Uncovered = 1 – Score

stop when it falls below your threshold

At most 5% of expected operational encounters involve insufficiently tested conditions.

a coverage claim, not a fault-absence claim

EVALUATION

Four datasets, seven baselines, including one built to stress our own method.

dataset	scenarios	failure rate	features
AsFault	5,638	45 %	7
DeepScenario	13,565	1.7 %	10
DeepCollision	8,605	60 %	18
Frost	5,294	7 %	9

Baselines: combinatorial coverage and two stricter variants, an empirical-frequency version, a diversity metric (TISA metrics), and two ablations of PWCC.

RESULT 1 · FALSE VERDICTS

The result I opened with, now with ours beside it.

combinatorial coverage

on the withheld splits

100 %

of unseen failures concealed · pairwise

99.7 % and 95.8 % at the higher interaction orders

PWCC

same splits, same suites

0

false verdicts · 0 failures concealed

across all thirty splits, at every interaction order

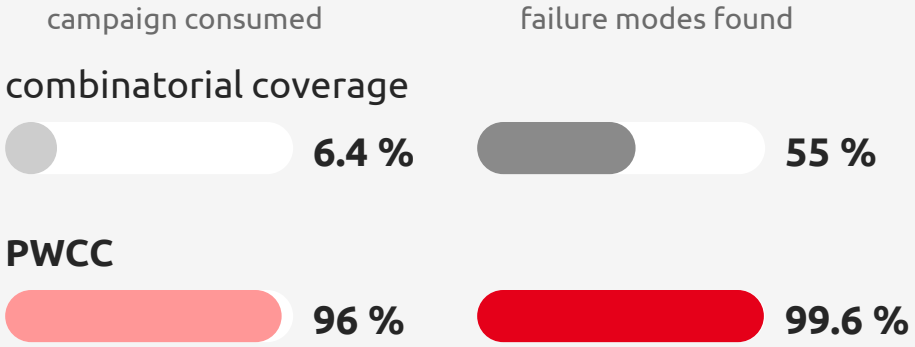
false verdict rate: how often a metric wrongly declares adequacy · concealment: the share of unseen failures that verdict hides

RESULT 2 · WHERE IT STOPS

Combinatorial coverage stopped after 6.4 % of the campaign with 198 of 437 failure modes undiscovered.

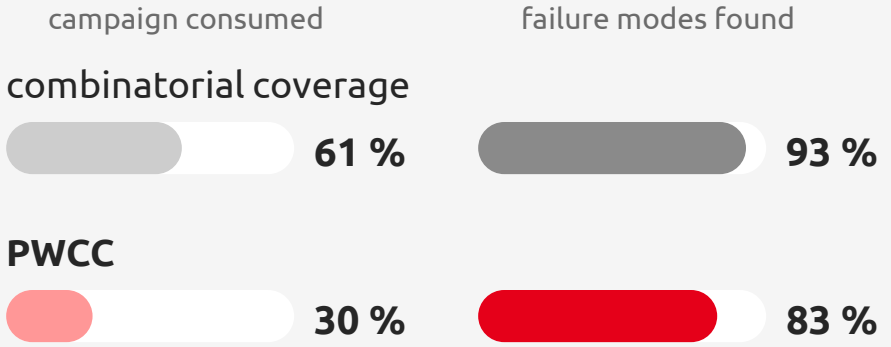
DeepScenario

failures rare: the gap is open



AsFault

failures dense: the gap has closed



later when the operational gap is still open, earlier when it has closed

RESULT 3 · ABLATION

Remove the learned weighting, and the metric certifies adequacy exactly where failures are densest.

PWCC with learned weights

-0.83

a low score predicts more failures

uniform weights

+0.82

a high score predicts more failures

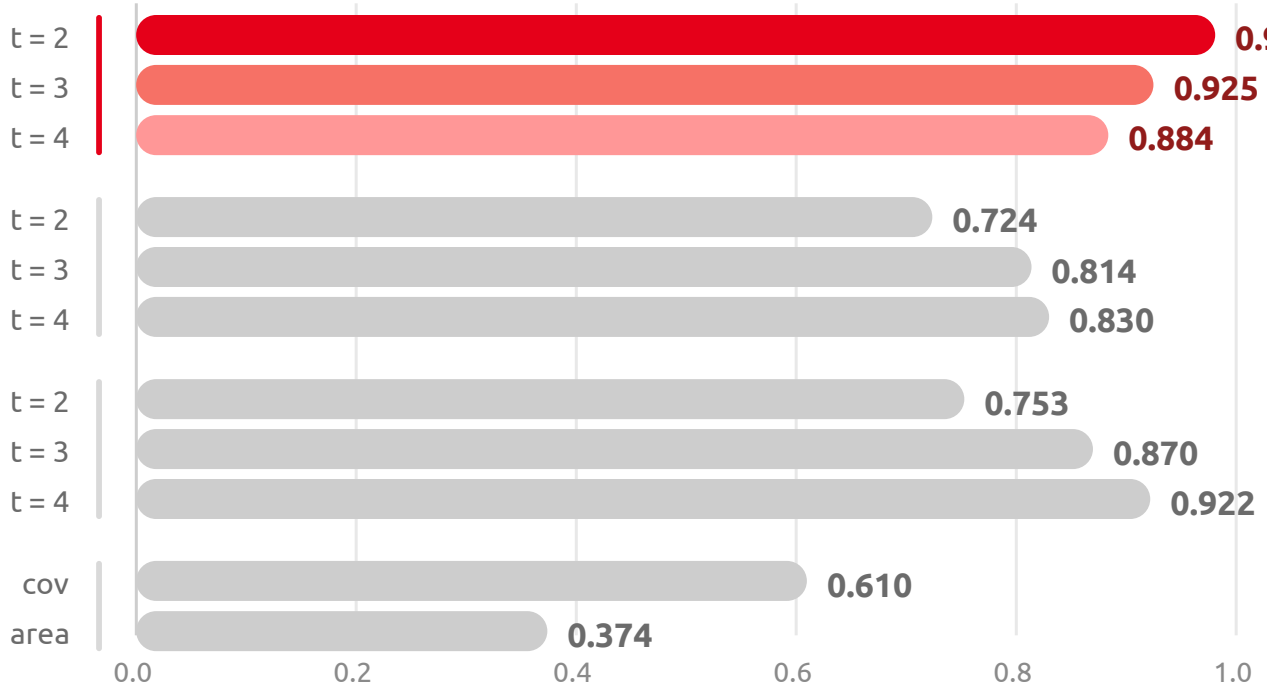
correlation between coverage score and failure density in the withheld region

same exposure criterion, same risk budget, same stopping rule. Only the weights changed.

RESULT 4 · FAULT DETECTION

Across the baselines, PWCC is the strongest predictor of fault detection.

SPEARMAN CORRELATION BETWEEN METRIC SCORE AND FAULT COUNT · DEEPCOLLISION · 30 SUITES OF 2,500 SCENARIOS



BEST OVERALL

0.981

PWCC at t = 2

+0.167

above the best combinatorial baseline, at a lower interaction order

where the operational distribution carries signal, probability weighting is what finds the faults

TISA's third metric, area occupied by failing scenarios, reaches −0.880 and is omitted here for scale.
At t = 4 every metric saturates on this dataset, so the comparison that matters is PWCC at t = 2 against the best lower-order baseline.

SCOPE

Two limits, stated.

1 **Conditional on the profiling data.**

If the profile does not match deployment, PWCC inherits that error.

2 **Frequency-weighted, not severity-weighted.**

Rare and catastrophic is a different question. PWCC sits alongside critical-scenario search, not in place of it.

CONCLUSION

Coverage becomes a stopping decision you can defend.

Existing metrics measure what has been tested. None of them measure how much of what the system will actually encounter is **still insufficiently tested**.

PWCC makes that quantity measurable, and turns coverage from a descriptive statistic into a stopping decision you can defend.

test adequacy | scenario-based testing | combinatorial interaction testing

artifact · all four datasets
doi.org/10.5281/zenodo.19181612

borhaneddine.hamadou@cea.fr